

Task 4: Optimized Step Selection (Encoder-Decoder)

1. Introduction

This task evaluates the trade-off between the number of diffusion steps (T) and generation quality. For the **Encoder-Decoder** variant, higher step counts significantly increase latency due to the iterative nature of the decoder forward pass.

Result: **T=4** was selected as the optimal production configuration for the Encoder-Decoder architecture.

2. Technical Performance at T=4

The T4 configuration utilizes a compressed diffusion schedule to minimize inference time while maintaining a baseline semantic coherence level of ~ 0.05 (for specific Sanskrit sloka tests).

👉 Implementation: [analysis/step_ablation.py](#)

Average Latency

~ 340 ms

(Local CPU / MPS)

Throughput

~ 235 Tokens/sec

(Batch Size = 1)

3. Conclusion

Similar to the cross-attention variant, the Encoder-Decoder model reaches a point of diminishing returns early in the diffusion process. Beyond $T=16$, the latency overhead of the decoder layers outweighs small gains in token probability calibration, justifying the T4 "Lean Generation" strategy.

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